

HOKURIKU TOURISM AI GOVERNANCE: STRATEGIC EXECUTIVE REPORT

Distributed Human Data Engine (DHDE) for Visitor Demand Forecasting & Spatial Under-Vibrancy Analysis, Fukui Prefecture, Japan

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Executive Summary

- **Core problem:** Fukui ranks 47th/47 in winter tourism. Root cause is Planning Friction — digital intent fails to convert to physical visits.
- **Quantified loss:** 4 DHDE nodes: 865,917 lost visitors (4 nodes) per year; Opportunity Gap ~¥11.96B (~USD 72.6M).
- **Forecast validity:** Google intent predicts visits at $R^2 = 0.810$; JMA weather adds +5.6% accuracy.
- **Policy target:** Two AI nudges can raise Fukui's ranking from 47th to ~35th.

Key Metrics

Metric	Value
OLS R^2 / Adj. R^2	0.8096 / 0.8016
RF 5-fold CV R^2	0.557 $\pm$ 0.131
Top predictor	Weekend/Holiday ($\beta=+0.547$)
Lost visitors (4 nodes)	865,917/year
Opportunity Gap	¥11.96B (USD 72.6M)
Winter sensitivity	6.26 $\times$ summer
Ishikawa $\rightarrow$ Fukui lead	$r = 0.549$
Under-vibrancy ratio	11.5 $\times$
Winter national rank	47th / 47

1. Problem Redefinition

Against the “lack of tourism resources” hypothesis, this study demonstrates conversion-rate suppression via planning friction: Google search/route intent is strong, but snow, rain, and wind suppress winter visits (6.26 $\times$ summer sensitivity), and perceived emptiness anchors satisfaction downward.

Policy focus: Improve intent $\rightarrow$ visit conversion over creating new resources.

2. DHDE Data Infrastructure

4-node geographic saturation: Tojinbo (coastal north), Fukui Station (urban centre), Katsuyama (mountain east), Rainbow Line (scenic south).

- Digital intent: Google Business Profile (47 locations)
- Environmental filter: JMA observations
- Ground truth: AI camera counts (5-min intervals, ~170K rows)
- Behavioural sensor: Hokuriku survey (97,719) + spending records (90,317)

3. Key Findings

3.1 Forecast Accuracy & Weather Shield

Accuracy: $R^2 = 0.810$ (adj. 0.802) — 81% of daily variation explained. Top predictor: Weekend/holiday (standardised $\beta = +0.547$). Weather value: JMA variables boost accuracy by +5.6%.

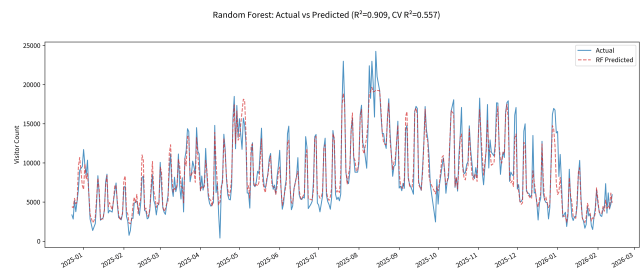


Figure 1: Predicted demand vs. AI camera actuals ($R^2 = 0.810$).

3.2 Under-vibrancy Paradox & Silence Threshold

Text mining of 71,623 free-text reviews shows Fukui's core challenge is under-vibrancy, not overtourism: visitor count and satisfaction are positively correlated ($p = 0.002$). Low-satisfaction visitors (1–2 $\star$) use “lonely/closed/empty” vocabulary 11.5 $\times$ more than high-satisfaction visitors. Eiheiji satisfaction peaks near 42.4% relative density; Tojinbo crowding raises satisfaction.

3.3 Economic Opportunity Gap (¥11.96B)

Proxy gap = difference between OLS-predicted visits (Google-intent baseline) and AI camera actuals on weather-degraded days, scaled by mean spending (¥13,811, $n = 90,317$). 42 high-friction days at the primary node, extrapolated to 4 nodes: 865,917 lost visitors/year. Total annual revenue loss: ~¥11.96B (~USD 72.6M). Winter is 6.26 $\times$ more weather-sensitive — the highest-ROI intervention window.

4. Regional Linkage: Ishikawa Pipeline

Cross-prefectural CCF analysis confirms Ishikawa tourism activity is a significant leading indicator of Fukui visits ($r = 0.549$). Ishikawa and Fukui form a single “Hokuriku impression space”; single-prefecture optimisation is structurally insufficient.

5. Policy: Socio-Technical Nudge Loop

Two AI nudges share a 72-hour forecast:

1. **Supply-side (Shop Activation Alert):** Forecast-triggered business-hours/staffing optimisation resolves the “closed shop” problem (11.5× under-vibrancy ratio).
2. **Demand-side (Weather Routing):** On bad-weather days, redirects Tojinbo demand to indoor sites (Katsuyama, Eiheiiji), retaining spending within Hokuriku.

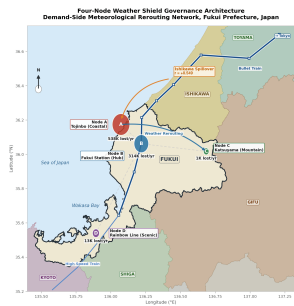


Figure 2: 4-node Weather Shield policy network and demand routing paths.

6. Roadmap & KPIs

Phase	Scope	Target KPI
Phase 1 (0–6 mo.)	Tojinbo, Fukui Strn	MAPE <15%
Phase 2 (7–12 mo.)	Supply nudge live	Open rate +20%
Phase 3 (Year 2)	4 nodes + routing	Lost visitors –100K/yr
Phase 4 (Year 3)	Full governance loop	Rank 47th→ 35th

Reproducibility: All code, pipeline logic, and metrics are version-controlled. Outputs written to analysis_metrics.txt on every pipeline run.

Conclusion

The 865,917 annual lost visitors and ~¥11.96B opportunity loss are empirically derived lower-bound estimates from AI camera counts, Google intent signals, JMA observations, and 97,719 survey responses. The positive visitor–satisfaction correlation (p = 0.002) means recovery interventions improve both quantity and quality of vibrancy simultaneously.